
Student Performance Prediction System

Problem statement ID: 27

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Project Phase-2 Report

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1. Problem Description

The Student Performance Prediction System is a machine learning based project that predicts student academic performance using different educational and behavioral factors such as study hours, attendance, internal score, previous results, sleep hours, and extra activities.

The main objective of the project is to help educational institutions identify students who may perform poorly and provide early support for improvement. The system processes the dataset, performs preprocessing and normalization, applies machine learning algorithms, and evaluates prediction accuracy.

The project uses classification techniques to predict pass/fail conditions and risk labels. Models such as Logistic Regression, Support Vector Machine (SVM), and Random Forest are used for prediction and analysis.

2. Architecture

Dataset Collection → Data Preprocessing → Feature Scaling → Feature Selection (PCA) → Model Training → Prediction → Evaluation Metrics

Steps Included:

- Data loading using Pandas
- Outlier detection using IQR method
- Data normalization using StandardScaler
- Feature selection using PCA
- Model training and testing
- Performance evaluation using Accuracy, Precision, Recall, F1-score, and Confusion Matrix

3. Models Used

Logistic Regression

Used for binary classification problems. It predicts whether a student will pass or fail based on input features.

Support Vector Machine (SVM)

A supervised learning model that separates classes using a decision boundary with maximum margin.

Random Forest

An ensemble learning method that combines multiple decision trees for better prediction accuracy.

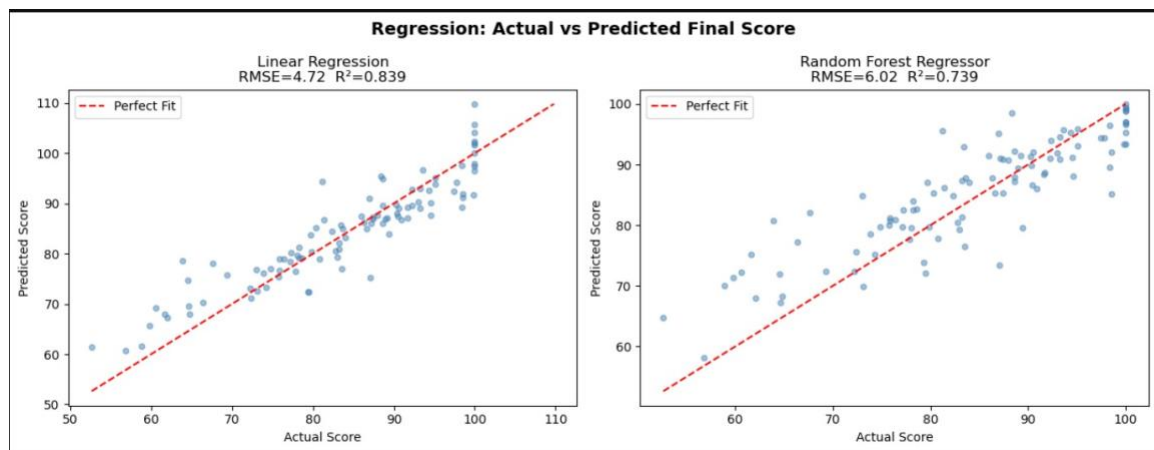
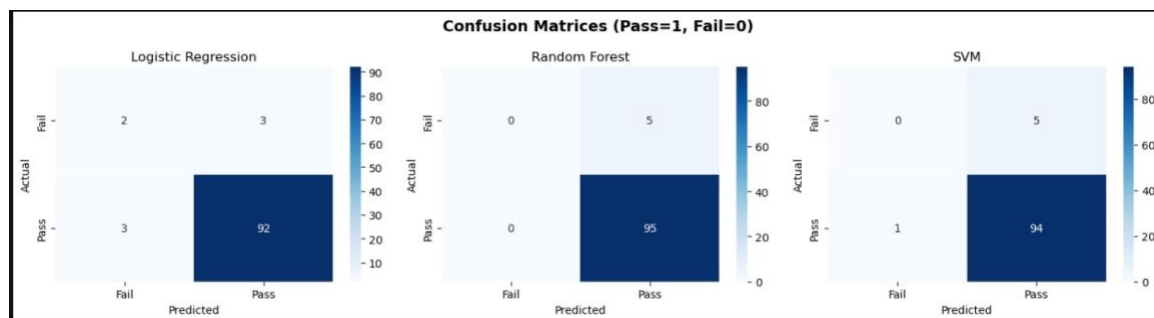
Linear Regression

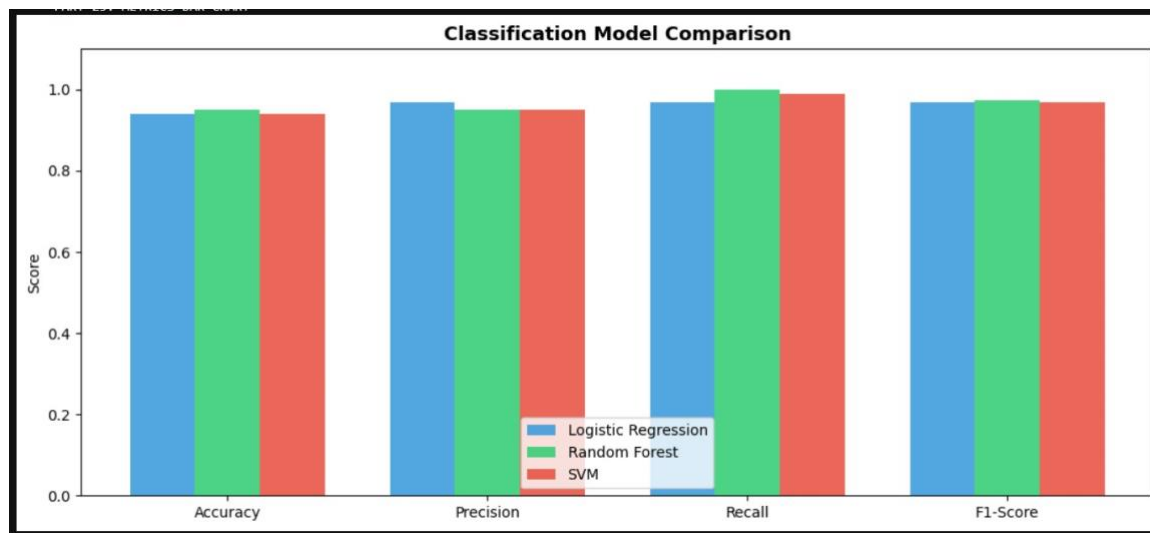
Used for predicting continuous output values such as final scores.

4. Evaluation Metrics Used

- **Accuracy:** Measures the overall correctness of predictions.
- **Precision:** Measures how many predicted positive values are actually correct.
- **Recall:** Measures how many actual positive values are correctly identified.
- **F1-Score:** Balances precision and recall.
- **Confusion Matrix:** Shows TP, TN, FP, and FN values for classification evaluation.
- **R2 Score:** Measures regression model performance.
- **Mean Squared Error:** Measures prediction error in regression tasks.

5. Preliminary Results and Visualizations





The dataset contains 500 student records with multiple performance-related attributes.

Important observations:

- No significant outliers were detected using the IQR method.
- Features were normalized successfully using StandardScaler.
- PCA was applied for feature selection and dimensionality reduction.
- The machine learning models achieved effective prediction performance for student risk and pass/fail classification.

6. Conclusion

The Student Performance Prediction System successfully demonstrates the use of machine learning techniques in educational analytics. The project helps predict student performance and identify risk levels using different academic and behavioral factors.

The system can assist teachers and institutions in improving student outcomes through early intervention and data-driven decision making.

7. References

- Python Documentation
- Scikit-learn Documentation
- Pandas Documentation
- NumPy Documentation

- [GeeksforGeeks](#)
- [Stack Overflow](#)